

Data Flow Diagram

AI-Assisted Smart Budget Planner

PocketSmart AI Data Flow

User → Web UI → FastAPI → Validation / Authentication → Planner Engine → SQLite / Optional Gemini → Plan Response → User

Element	Role
External Entity: User	Provides registration, login and planner inputs; receives plan results.
Process 1: Authentication	Registers users, validates credentials, manages session/JWT.
Process 2: Planner Request	Validates Home/Party/Jewelry request data.
Process 3: Shared Planning Engine	Uses catalog and deterministic packing to allocate affordable whole units.
Data Store: SQLite	Stores users and saved plans/history.
External Service: Gemini	Optional AI commentary; does not control arithmetic.
Output	Itemized plan, total, remaining budget, notes, insight and search links.

Jewelry image flow: optional JPEG/PNG/WebP (≤3 MB) → signature validation → optional Gemini analysis for commentary; the image is not stored and does not affect pricing.